

Crabs on camera: project documentation

Contents

1	Background	5
1.1	Project and repository aims	5
1.2	About the repository	5
1.3	Before running the scripts	6
2	Create a crab detector	9
2.1	From raw annotations to a YOLO model	9
2.2	Training the crab detector	12
3	Use a crab detector	17
3.1	Background	17
3.2	Before running the detector - independent test set	18
3.3	Running the detector	19
3.4	Post-processing-pipeline	21
4	Glossary	25

Chapter 1

Background

1.1 Project and repository aims

The basis of this project is rooted in the work done in the masters thesis by Brøten [2026]. The aim of the project was to improve monitoring of the data-limited brown crab (*Cancer pagurus*) stock in Norway using machine learning and underwater video surveys. With the scripts and methods applied in this repository you can: 1. create your own detector for crab; 2. apply either the pretrained crab detector, or your own.

The aim of this repository is not to push the boundaries of machine learning or showcase new methods, rather to make these methods more approachable for marine scientists working in fisheries applications. A glossary is found in Chapter 4.

AI tools were used throughout this repository to help generate code, fix syntax, and clean up errors.

1.2 About the repository

This repository uses the same code, functions, and packages developed for the masters thesis, though some functions have been simplified here to make the pipeline easier to reproduce.

The structure of the repository is as follows:

```
crabs-on-camera/  
  create-a-crab-detector/  
    data/  
      annotations/  
        test_subset/                                # example dataset shipped with the repo
```

```

        annotations/
            instances_default.json    # CVAT/COCO export: labels + crab outlines
            images/default/           # the video frames those annotations belong to
script/
    1-raw_to_yolo.ipynb              # converts raw annotations into a YOLO-readable format
    2-train.ipynb                   # trains the YOLO model
    setup-scripts-optional/
        annotation_stats.ipynb      # summary statistics on the annotation data
        annotation_format/          # template CVAT label schema, for annotating images
        extract_frames/             # script for collecting frames for annotation
use-a-crab-detector/
    model/
        pretrained-crab-detector/   # the trained model from the thesis (best.pt)
    data/
        videos/                     # your own BRUV footage to run the detector on (no annotations)
        predictions/                 # raw detections and post-processing output, generated from videos
            annotated_videos/        # optional annotated output videos, if SAVE_VIDEOS is True
        test_predictions/            # the independent test set: annotations, images, predictions
    script/
        1-get-ready-to-use-a-crab-detector.ipynb # compares model predictions to truth
        2-use-a-detector.ipynb          # runs the model on unseen material
        3-interpret-raw-detections.ipynb  # interprets raw predictions
        4-post-processing-pipeline.ipynb  # removal of likely false detections
    docs/                             # this documentation
    README.md

```

Everything needed to reproduce the pipeline lives in three places: `data/` holds a small data set you can run the scripts against immediately, `script/` holds the actual pipeline as a series of notebooks, and `docs/` is the code documentation.

1.3 Before running the scripts

1.3.1 Prerequisites

To run the scripts, clone a copy of the repository to a local folder or server, depending on the amounts of data is going to be processed. A server with GPU access is recommended if you plan to train on a large dataset, the small example datasets shipped with this repository can be run locally without one. To run the scripts multiple IDEs are possible, such as VS Studio Code. Python needs to be downloaded, the scripts are primarily jupyter notebooks, which follow a similar structure as .rmd documents with chunks. At the start of both initial scripts in the repository, checks for missing packages and installs them. All packages required for the script to run is freely available and open-source.

To identify and quantify crabs in the videos, this project uses the object detection framework “You Only Look Once” (YOLO), specifically version 8 [Jocher

et al., 2023], which builds on the original YOLO architecture introduced by Redmon et al. [2016]. Several object detection frameworks, including newer YOLO versions, could have been used instead. YOLOv8 was chosen because it is well documented and its common issues are well known, which makes it easier to troubleshoot for beginners in machine learning.

A glossary for all terms can be found in Chapter 4.

1.3.2 Data

The repository ships two example datasets so the scripts can be run and verified without applying new data. The both of the datasets are from Institute of Marine Research inshore gill-and fyke-net survey conducted in 2025 & 2026. The datasets are image frames collected from Baited Remote Underwater Videos (BRUVs), alongside annotations which are exported from CVAT, as a .JSON file, which is a plain text file storing structured data, in this case annotations, and such as categories (here these are species groups). The images are labelled with multiple identifiers to make analysis and stratified sampling later simpler.

E.g., **NO_2025_3007_GarnRuse_Lovund_LO3_RA1_L9_frame_9**, is collected in Norway, in 2025 on the 30.07 on the Garn and Ruse survey (gill-and fyke-net), the location name is Lovund, and its station number is LO3, and the BRUV-rig used is L9, and this is video number 9 from the left camera (L9), and is frame number 9. So if you are applying another type of video-survey or other naming logic please be mindful, though its possible to turn off this stratified sampling in the functions.

To collect frames for annotations, an example script is in the repo (`script/setup-scripts-optional/extract_frames`). Likewise a .JSON file for the structure applied in this annotation process can be found in (`script/setup-scripts-optional/annotation_format`). For more in depth information about the sampling methods and areas, annotation choices and a discussion of methods can be found in Brøten [2026].

Chapter 2

Create a crab detector

The folder `create-a-crab-detector` contains scripts and datasets which makes it possible to go from raw annotations to a trained YOLO-model. The first script (`1-raw-to-yolo.ipnb`) takes raw CVAT annotations and turns them into a dataset ready for training an YOLO object detection model: RLE mask are converted into bounding boxes, and the data is split into five stratified folds for cross-validation.

If you're applying this dataset to other than the shipped example, keep in mind (but not limited to); file paths: `CONFIG` needs to point at your own annotation JSON and image folder, and stratified sampling for the k-folds, this assumes your images can be grouped by BRUV station and labeled crab-present/empty; if that doesn't apply to your data, this step may need adjusting or turning off (see Chunk 9).

By the end of the script, each image should have a matching `.txt` file listing its bounding box annotations (or an empty file, if there's no crab in it), organized into five train/validation folds ready for object detection training.

2.1 From raw annotations to a YOLO model

1. Setup and loading data (Chunk 0 – 3)

These chunks handle setup: installing any missing Python packages, loading the required libraries, and locating your dataset. They should run without errors, if Chunk 3 can't find your annotation file or image folder, double check the paths.

2. Converting masks to bounding boxes (Chunk 4)

RLE masks are precise, but training directly on them takes more compute time and is really built for image segmentation, not simple object detection. Since

the goal here is just to find and count crabs (not trace their exact outline), converting to bounding boxes is the better trade-off as they’re faster to train on, and sufficient for detection. RLE masks are faster to draw as the annotation program utilized for the example datasets, Computer Vision Annotation Tool (CVAT; Sekachev, 2020) applies Segment Anything Model 2 from META [Ravi et al., 2024] which makes the annotation process much faster than drawing traditional bounding boxes.

This chunk converts each RLE mask into a tightly-fitted bounding box, and normalizes the box coordinates to the 0—1 range YOLO expects. Two filters are applied at the same time: annotations for any species other than decapod: crab are discarded, and overlapping duplicate boxes (drawn twice for the same crab) are reduced to one.

3. Linking images to annotations (Chunk 5)

Links each annotation to its actual image file, matching by filename, again make sure image filenames match the identifiers in the annotation JSON exactly, any mismatches get reported as “unmatched,” usually caused by a typo, a renamed file, or an image missing from the folder.

Note: frames with no crab in them (negative examples) are deliberately kept in the dataset rather than discarded, training on empty frames as well as crab-present ones has been shown to improve performance [Sorrenti et al., 2025]. These get an empty `.txt` label file later, once the data is exported (Chunk 9).

4. Visual quality checks (Chunk 6 – 7)

These chunks are a visual quality check, not a processing step, and nothing here changes the data. Chunk 6 displays the first annotated image with its converted bounding box overlaid: every crab should have a red box tightly wrapping it, with no gaps or extra boxes.

Chunk 7 places the original RLE mask and the converted bounding box for the same crab side by side, to visually confirm the conversion in Chunk 4 didn’t distort or misplace anything before relying on it for training.

5. Location and station helpers (Chunk 8)

Filenames encode which location and BRUV station each image came from, but that information isn’t stored as a separate field anywhere, it has to be pulled out of the filename itself. This chunk does that: `extract_location` reads out the site name (e.g. “Lovund”), and `extract_bruv_id` reads out the specific station ID (e.g. “LO3”). It also prints a quick summary: how many images come from each location, and how many of those are crab-present versus empty, worth a glance to check whether the dataset is skewed toward a few sites, since that affects how balanced the folds can be.

If applying another dataset with a different naming logic, these two functions will need to be adjusted to match the naming pattern (see Section 1.3.2).

6. Splitting into stratified, grouped folds (Chunk 9)

This is the core of the export step: it splits the linked dataset into five folds for cross-validation, applying two rules at once.

Grouped: every image from the same BRUV station stays entirely in either training or validation within a fold, never split between the two. If the same station showed up in both, the model could effectively be tested on footage from a site it had already trained on, making it look like it's performing better than it actually would on a genuinely new site.

Stratified: within that constraint, each fold's train/validation split keeps a similar proportion of crab-present vs. empty frames as the full dataset, so no fold accidentally ends up training almost entirely on empty frames.

The function has a `stratify` option: `True` (the default) gives the behavior above; `False` skips the crab/empty balancing and keeps only the station grouping. Turning it off makes sense if your dataset is already fairly balanced, or if the crab/empty ratio varying between folds isn't a concern for you, station grouping still applies either way, so there's no risk of leakage regardless of this setting.

For each fold, this chunk writes the images and label files into `images/train`, `images/val`, `labels/train`, `labels/val`, along with a `dataset.yaml`.

YAML is a plain-text format for storing settings, written as simple, human-readable lines rather than code. YOLO's training tool reads `dataset.yaml` to know where to find the images and labels and what classes to expect. Each fold's file specifies: `path` (that fold's root folder), `train` and `val` (the image subfolders), `nc` (number of classes - 1, since this project only detects crabs), and `names` (mapping each class number to its name, e.g. 0: `crab`).

7. Running the export (Chunk 10)

This chunk runs the function from Chunk 9 on your dataset. By default it creates five folds under `data/kfold/`, each structured as:

```
data/kfold/
  fold1/
    images/train/
    images/val/
    labels/train/
    labels/val/
    dataset.yaml
  fold2/ ... fold5/      # same structure as fold1
  master_dataset.yaml
```

`fold_summary.csv`

along with a `master_dataset.yaml`, a small reference file at the top level listing the class setup (`nc` and `names`, same as each fold’s file) and a comment reminding you how to point training at a specific fold (e.g. `yolo train data=fold1/dataset.yaml ...`), and a `fold_summary.csv` with the actual per-fold stats mentioned above.

Once this runs cleanly, no station-overlap warnings, and a crab/empty ratio that looks reasonable in the fold summary, the data is ready for training in `train.ipynb`.

A caveat is that with this limited dataset of only ten stations, we are not expecting the folds to be perfectly balanced, and it is expected that the folds that look more unbalanced will perform worse than the others. Now the dataset is ready for training!

2.2 Training the crab detector

This script (`train.ipynb`) takes the five-fold dataset produced by `1-raw_to_yolo.ipynb` and trains a YOLOv8 object detector on it, one model per fold, so performance can be evaluated across several independent train/validation splits rather than trusting a single one.

1. Setup (Chunk 1)

This chunk locates the repository (the same `find_repo_root()` approach used in `1-raw_to_yolo.ipynb`, so paths work the same whether its running on a laptop or a server), sets up where training output gets written, and checks that a GPU is available. Training a model like this on a CPU is impractically slow, so this check is worth paying attention to: if `CUDA available: False` prints here, stop and sort out the GPU setup before continuing rather than letting a full training run crawl for days.

2. Checking the folds (Chunk 2)

Before spending hours training, this chunk verifies every fold from `1-raw_to_yolo.ipynb` is actually complete: that each fold has a readable `dataset.yaml`, and that the number of images and label files line up in both the train and validation sets. This is a safety gate, not a formality, training on a broken or incomplete fold wastes real compute time, so it’s worth letting this fail loudly here rather than partway through a long run. Chunk 5 (the real training) refuses to start if this check doesn’t pass.

3. Hyperparameters (Chunk 3)

This chunk sets out every training parameter in one place, so the whole run is reproducible from a single configuration. Rather than explaining every individual number, it's easier to think of them in three groups:

Model and image handling. `yolov8s.pt` is the pretrained starting point the model builds on rather than learning from scratch, you can freely try other sizes of YOLOv8 or other versions by changing them out here. Images are resized to 960 pixels and processed in batches of 16, larger images and batches use more GPU memory, but a larger image size in particular can help the model pick out small objects like crabs against a wide frame.

How the model learns. The optimizer, learning rate, and warmup settings control how aggressively the model adjusts itself while training, and how that pace changes over the course of the run (a “cosine” schedule, `cos_lr`, means the learning rate eases off smoothly toward the end rather than dropping abruptly). These values were carried over from what worked in the original thesis training runs, rather than re-tuned from scratch here.

Augmentation. During training, images are randomly rotated, shifted, scaled, and mirrored slightly before being shown to the model. The intent isn't to create unrealistic images, it's to stop the model from memorizing the exact footage it was trained on, so it generalizes better to new footage. A couple of these are deliberately constrained for this dataset: `flipud=0` means images are never flipped upside-down (a crab is never upside-down in real footage, so training on flipped versions would teach the model the wrong thing), and `single_cls=True` reflects that there's only one class to detect here (crab), so the model doesn't spend effort distinguishing between species that don't apply.

Two epoch counts are set here: `test_epochs` (5) for the quick sanity check in Chunk 4, and `epochs` (100) for the real run in Chunk 5, 100 was chosen because performance had already converged by that point in the original thesis training.

4. Test run (Chunk 4)

This runs a handful of epochs on one fold using the exact same settings as the full run, purely to confirm the pipeline works end-to-end, paths resolve, the GPU is used correctly, and training completes without errors before committing to a run that takes hours. The actual metrics from this run aren't meaningful and shouldn't be read as a preview of model quality: five epochs isn't enough for the model to properly learn, and results can look noisy or even get temporarily worse epoch to epoch. That's expected here, not a warning sign.

5. Full training (Chunk 5)

This trains one full model per fold, using the hyperparameters from Chunk 3. Each fold's best-performing checkpoint is saved automatically, along with a

checkpoint every 25 epochs during training in case a run needs to be resumed. There’s a short pause between folds to let the GPU cool down. The current server utilized is NVIDIA RTX A6000, which used around 3-4 hours for the entire training run, whilst training the full dataset used in the thesis took around 16–18 hours.

6. Collecting and interpreting results (Chunks 6 – 8)

Training five separate models means five separate sets of results, so these chunks pull them together: loading each fold’s training log, plotting the key metrics (mAP, precision, recall) side by side across folds, and then computing a summary, both the final numbers from training and a fresh, explicit validation pass on each fold’s saved model, as a double-check against the training log alone.

The reason to look across all folds rather than picking one is the same logic behind K-fold cross-validation in general: any single fold could happen to get an easy or a hard validation split by chance, so the story that matters is the overall pattern, not one number in isolation.

Interpretation

One limitation worth being upfront about: this dataset has only 10 BRUV stations, and because stations are kept fully separate between training and validation (to avoid the model being tested on footage it effectively already trained on – see the K-fold export step in `1-raw_to_yolo.ipynb`), each fold’s validation set ends up being just 1-3 stations. That’s a small, uneven sample, and fold-to-fold variation in these results should be read as a consequence of the dataset’s size, not a flaw in the training method. One fold in particular had very few crab-positive images in its validation set purely by chance, and its numbers should be treated as unreliable rather than averaged in on equal footing with the others.

To evaluate object detection model performance, standard object detection metrics were used: true positives, false positives, false negatives, mean average precision (mAP50), precision, recall, and F1 (Table 2.1). After training, each model was validated on the validation dataset. The model predicts the class and location of objects in each frame with bounding boxes. To assign these metrics, predictions were ranked by confidence and matched to the manually annotated bounding boxes from the validation dataset, with each annotation matched to at most one prediction. A predicted box was counted as a true positive (TP) if it overlapped an annotation of the same class sufficiently, assessed using intersection-over-union (IoU); unmatched predictions were counted as false positives (FP) and unmatched annotations as false negatives (FN) (a confusion matrix showing the distribution of all these can be found in `test_runs/models/fold1_yolov8s_100epochs`). IoU is calculated as the area of overlap between the predicted and annotated box divided by the area of their union, giving a score between 0 and 1 [Biscarini and Nazzicari, 2026].

Table 2.1: Standard performance metrics for assessing object detection models. A detection is counted as correct (TP) if the predicted bounding box overlaps with an annotated box by at least 50% ($\text{IoU} \geq 0.5$). Metrics range from 0 to 1, where 1 indicates perfect performance.

Metric	Description
True positive (TP)	Correct detections: predicted bounding box matches an annotated box.
False positive (FP)	Model predicts a bounding box where there is no object.
False negative (FN)	An object is present but the model produces no matching prediction.
Precision (0–1)	Ratio of correct positive predictions to all positive predictions, measures detection quality.
Recall (0–1)	Ratio of correct positive predictions to all actual positives, measures the model's ability to find all objects.
F1 (0–1)	Harmonic mean of precision and recall, balances avoiding false detections with finding all objects.
Mean average precision (mAP50)	Average of precision scores across recall levels, summarized across all classes (COCO standard).

An IoU threshold of 0.5 was applied, meaning a prediction had to overlap at least 50% with the annotated box to be counted as correct, following standard object detection practice.

Aside for interpreting the performance metrics, its worth checking out the visual validation confirmations. In the folder `test_runs/models/fold1_yolov8s_100epochs`, or any other fold. Compare the `val_batch_labels.jpg` with `val_batch_pred.jpg`, this compares the actual annotated images and the predicted annotations from the model to see the confidence scores and what the model catches/misses. If the behavior seems wrong, there is a plethora of tweaks possible to improve model behavior which is easily available online or with the help of LLMs.

Chapter 3

Use a crab detector

3.1 Background

The aim of this leg of the repository is to go from unseen data to assessment ready data, where unseen video material is the input and the output is the relative abundance indices MaxN and MeanN per BRUV/video station obtained with either the pretrained crab detector, or another model.

MaxN is the single image frame within a deployment with the highest count of individuals per deployment, whereas MeanN is the mean number of individual observed over a series of image frames [Whitmarsh et al., 2017]. MaxN is the most commonly applied metric of estimating abundances in video-based surveys, and its conservative with its aim to reduce the likelihood of double counting individuals.

The pretrained crab detector developed in the thesis is included in this repository, in the folder `use-a-crab-detector/model/pretrained-crab-detector`. This model is not yet plug and play, as documented in the thesis, and in the literature more broadly, YOLO models in particular struggle to generalize: when the input material shifts away from the domain it was trained on, performance drops [Lürig et al., 2021]. Specifically, this model was trained to detect brown crab in BRUVs placed in nearshore coastal areas between 62°N and 68°N. When applied outside this scope, whether due to a different video collection method or a different environment, it's going to struggle (see the FjordFish test in Brøten [2026]).

If you'd still like to apply this detector given these caveats, or apply a new (and likely better) model, some manual work is still needed. To interpret the model's predictions, we need some understanding of its generalizability. One important step is building an independent test set: annotate a small subset of unseen data, let the model predict on it, then compare the model's predictions against the

human-annotated ground truth.

Recommended running order: 1. 1-get-ready-to-use-a-crab-detector 2. 2-use-a-detector 3. 3-interpret-raw-detections 4. 4-post-processing-pipeline

3.2 Before running the detector - independent test set

Often described as the gold standard for evaluating a trained models performance, and applied after a model is fully trained to assess the ability of the model to generalize, and its ability to detect general features rather than memorizing its training data [Biscarini and Nazzicari, 2026]. In this instance, its purpose is rather to evaluate the models “compatibility” with the data its going to predict upon. The model will predict on a small annotated dataset, and the predictions will be compared to the annotations. All this is done in the script 1-get-ready-to-use-a-crab-detector.

3.2.1 Setup an collecting predictions (Chunk 0 – 3)

To start, you will need a fully annotated dataset and should be around 10 – 25% of the available data. Alongside this repository is a small dataset which can be applied in the following scripts to test the methods. Likewise to the data processing pipeline performed in 1-raw_to_yolo the annotations are converted from RLE-masks to bounding boxes, and only the crab annotations are kept. The scrips logic for grouping is still based on the naming convention as previously mentioned, if the dataset uses another naming convention the code needs to be readjusted. The model will then predict on the dataset assigned, these are saved in use-a-crab-detector/data/test_predictions/predictions.json.

3.2.2 Comparing model predictions to human ground truth (Chunk 4 – 6)

First the per stations maxN, meanN and total crab count from the human-annotated test set is collected, this is the ground truth the predictions are compared against, then the same is done for the models predictions. Duplicated bounding boxes (two bounding boxes overlapping) are removed, and every image at a station is counted even when the model found nothing. As the model predicts it assigns a confidence score to the prediction from 0 – 1 based on how sure it is, the confidence threshold for predictions can be adjusted. High confidence thresholds keeps only the models most certain predictions (fewer false detections, but real crabs might be missed), whereas low confidence thresholds introduce introduces noise but retains more predictions. Because there is no single correct threshold, the script evaluates several thresholds (none, 0.5 and 0.75 by default) so you can see the trade-off rather than guessing.

The script saves two things: `threshold_summary.csv`, with one row per threshold giving the total crab count and average maxN/meanN for both the human annotator alongside three bar charts (`total_crabs_by_station.png`, `maxN_by_station.png`, `meanN_by_station.png`) comparing the human truth against the model at each threshold.

In the thesis, this comparison highlighted that neither a low or a high threshold was able to give an accurate MaxN prediction and highlighting the need for a more flexible solution for interpreting the detentions, this is covered in the post-processing-pipeline. To further understand what features drove the detector to make a prediction, a Grad-CAM analysis can be performed, which produces a heatmap highlighting important areas for detection.

3.3 Running the detector

3.3.1 Applying the detector to new footage

Once the crab-detector returned acceptable results on the independent testset, `2-use.-crab-detector` runs the model across a full folder of new BRUV videos. Point it (change the path) at the folder containing the desired footage and desired model, its also possible to change the image size, hyperparameters (though these should be close to the initial training setup) and confidence threshold. The script makes the detector predict roughly every second (matching the extraction of training data), but this can be changed, runs the detector on these frames and collects every detection into a single `predictions.json` file.

The confidence threshold is set to be deliberately low (0.1), hence the script is not deciding what counts as a “real” crab, just recording every prediction. The actual filtering, so choosing a confidence threshold and removing likely false detection happens in the next steps. By default this script only saves the detections and not annotated videos, to activate this there is a toggle in the scripts configuration to save a copy of each video with the predicted videos drawn in, off is the default since it adds significant disk use and processing time.

3.3.2 Checking what the model found

Before doing any real filtering, it is worth looking at what the detector actually predicted. `3-interpret_detections` reads the `predictions.json` file and reports the total number of detections, how many frames had at least one and how many detections fall above a couple confidence cutoffs (0.5 and 0.75 by default). It plots a histogram of the confidence score of every detection as a sanity check, and should (if you are applying the pretrained model) have a bimodal structure, with a high peak withing the lower end of confidence scores, and a second smaller peak with the higher scores (Figure 3.1).

In the last chunk it applies a soft tracker to display



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Essentially it shows how many consecutive frames a bounding box was present, and with the current crab detector used in the repo there will likely be a big amount of detections appearing for one frame and then disappearing. Since crab are slow movers (or at least slower than moist fish), we can assume that they stay in frame for longer than one second, and three seconds. Hence, all detections that are present for less than three seconds, we can assume were false. Likewise, crabs do move, so detections appearing in the same position for over 500 frames can be assumed to be false - both of these are removed in the post-processing pipeline (Figure 3.2).

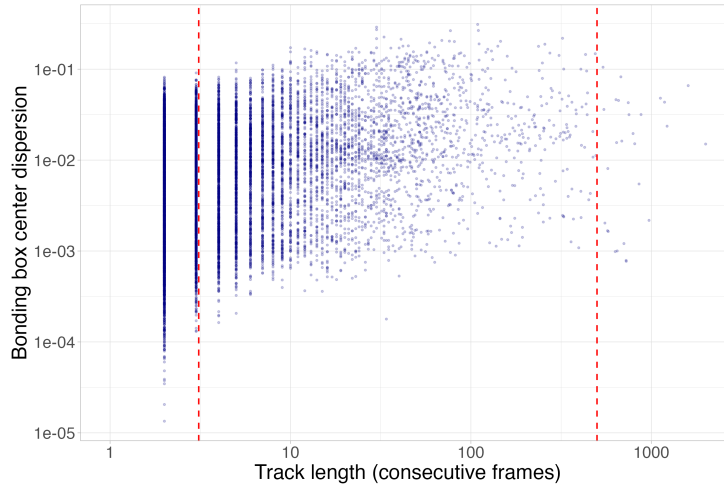


Figure 3.2: Relationship between track length (number of consecutive frames in which a detection appears) and bounding box center dispersion for crab detections generated by the crab detector in Brøten (2026). Dashed lines indicate track length thresholds applied during post-processing of the predictions (3-500 frames).

3.4 Post-processing-pipeline

This step is optional, but shows the workflow for removing likely false detections whilst maximizing the amount of true crab predictions beyond what confidence thresholding alone could achieve in **4-post-processing-pipeline**

applied in thesis. The idea behind was based on visual observation of the behavior of the crab in the model aswell as the models predictions behavior. To maximize true crab detections beyond what confidence thresholding alone could achieve, a post-processing pipeline was developed based on visual inspection of model predictions and known crab behavior in the BRUV footage.

Occasionally the model produced many overlapping bounding boxes for the

same individual crab. This is a common issue with object detection models, and the removal of these are called sequential Non-Maximum Supression, the same technique is applied here and also seen in a study for counting for fish by Connolly et al. [2021].

To address this, deduplication filters were applied within each individual frame with IoU. Pairwise IoU were calculated between all bounding boxes from model inference. Based on comparisons of multiple IoU thresholds, an IoU threshold of 0.3 was found to remove most overlapping bounding boxes, lower than 0.5 used in model evaluation, as the goal was to remove duplicate detections rather than assess accuracy. Only the largest bounding box (by area) was kept under the assumption that it was more likely to contain the entire crab (Figure 3.3).



Figure 3.3: A random frame from inference on BRUV footage from the 2024 gill-and fyke-net survey, where there are two detections (yellow boxes), and their bounding boxes overlap more than 30% ($\text{IoU} \geq 0.3$) over the same crab. After applying a post-processing pipeline to remove noise and unlikely predictions of crab, only the largest box of the two is kept.

After deduplication, all predictions with a confidence of ≥ 0.3 were kept, based on visual inspections that true crab predictions were rarely below this threshold.

The crab's behavior in the BRUV footage differs from fish, they moved slower and stayed in the frame for extended periods. To exploit this behavior, a tracking-based filtering approach was implemented. Detections were linked across consecutive frames into tracks based on the spatial proximity of bounding box centers and bounding box overlap in consecutive frames. Tracks were allowed to persist across short frame gaps (± 2 frames) to account for missed detections. Tracks with a duration of fewer than 3 frames, was assumed to likely represent false detections, and was removed. Conversely, tracks persisting for

over 500 frames were also removed as these typically corresponded to static false detections such as substrate features misidentified across the deployment. The predictions that were left after the post-processing pipeline were saved for further analysis.

The post-processing pipeline therefore consisted of four steps: 1) bounding box deduplication using IoU filtering, 2) confidence thresholding, 3) temporal tracking of detections, and 4) removal of short.

The two last chunks collect a .csv file showing amongst other, the station, raw maxN, filtered maxN, and the frames it appeared. For further analysis doing a e.g. `left_join` with the station for the metadata about the location for further analysis. In rFramewise comparison of the raw maxNs and the filtered maxN for visual comparison. The main takeaway from the visual confirmation should be that the filtered predictions generally look better, but there will most likely be errors.

Chapter 4

Glossary

- **Object detection:** a computer vision task where a model finds where objects are in an image and identifies what they are — in this project, locating and identifying crabs in video frames.
- **YOLO:** “You Only Look Once” object detection framework by Redmon et al. [2016], commonly used for identification and classification of marine species. YOLO combines object localization and classification in one step, predicting bounding boxes and class labels directly, enabling processing of large volumes of data and facilitating for species identification, abundance counts and size estimations for fisheries applications.
- **CVAT:** Open-source program Computer Vision Annotation Tool (CVAT; Sekachev et al., 2020) for annotation.
- **Bounding box:** a rectangular region defined by four coordinates that encloses a specific object within an image or frame.
- **RLE mask** (Run-Length Encoded mask): a pixel-by-pixel outline of an object’s exact shape, stored in a compressed format — this is how the original CVAT annotations are drawn, before being converted to bounding boxes.
- **Annotation:** a label marking an object within an image — in CVAT, drawn as a mask or box around each crab.
- **Confidence threshold:** when a machine learning model predicts, it assigns the prediction a score from 0–1 based on how sure it is about the detection, where 1 is most certain.
- **IoU** (Intersection over Union): a measure of how much two boxes overlap, used both to remove duplicate boxes during annotation cleanup and later as an evaluation metric.
- **Fold:** one of the K train/validation splits created during K-fold cross-validation.
- **K-fold cross-validation:** splitting the dataset into K equal-sized groups (“folds”). In each round, one fold is held out for validation while the

model trains on the rest, repeated K times so every image gets used for both training and validation, just never in the same round. Example: with 5-fold cross-validation, the model is trained and validated 5 separate times, each on a different split, so results don't depend on one lucky (or unlucky) train/validation split.

- **Stratified sampling:** splitting data so each subset keeps roughly the same proportion of a key category as the full dataset. Example: if 30% of all images contain a crab, stratified sampling makes sure each fold's train and validation sets are also close to 30% crab-present, instead of one fold accidentally ending up almost entirely empty frames.
- **Station leakage:** when images from the same BRUV station appear in both the training and validation sets, letting the model partly be tested on footage from a site it already trained on, this project's K -fold export explicitly prevents it by grouping all images from a station together.
- **Negative example / empty frame:** a frame with no crab in it, deliberately kept in the dataset (rather than discarded) because training on empty frames alongside crab-present ones has been shown to improve performance.
- **Ground truth:** the human-annotated "correct answer" (the CVAT masks/boxes) that a model's predictions are compared against.
- **YAML:** a plain-text file format for configuration settings, structured by indentation rather than punctuation: used here to tell the training script where the data lives and what classes to detect.
- **MaxN:** MaxN is a relative abundance index, which has the intent to reflect local density while avoiding double counting of individuals.

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